

Mayank Raj

vmayank0123@gmail.com | linkedin.com/in/mayankraj17 | github.com/Raj-Mayank2 | 9304618177

PROFESSIONAL SUMMARY

Final-year B.Tech **Information Technology** student with hands-on academic and project-based experience in full-stack development. Skilled in **C++**, **Python**, **Node.js**, and **MongoDB**. Built REST APIs and scalable web applications through projects and hackathons. Strong foundation in **Object-Oriented Programming (OOP)**, **Operating Systems**, **Database Management Systems (DBMS)**, and **problem-solving**.

EDUCATION

Maulana Abul Kalam Azad University of Technology, B.Tech in Information Technology June 2022 – Present

- CGPA: 9.18/10.0 (till 7th semester)

St.Michael's High School, Higher Secondary

May 2019 – 2021

- Percentage: 96.2/100

DAV Centenary Public School, Class 10th

May 2018 – 2019

- Percentage: 98/100

TECHNICAL SKILLS

Languages: C++, C, Python

Web and Backend: Node.js, React.js, Express.js, REST APIs, JWT-based Authentication

Databases and Tools: MySQL, MongoDB | Git, GitHub, Postman, VS Code

Core Subjects: Operating Systems (OS), Data Structures and Algorithms (DSA), Database Management Systems (DBMS), Object-Oriented Programming (OOP)

PROJECTS

Aayojan.ai

[Link](#)

- **Awarded 1st Place at the Metamorph 1.0 Hackathon** for developing an innovative AI-powered event management platform offering seamless user experience and practical automation features.
- Developed Aayojan.ai, a full-stack event and community management system using **Gemini AI models** to automate financial reporting, sponsor letter generation, marketing emails, and digital ticketing with secure QR-based access.
- Tools Used: React.js, Leaflet.js, MongoDB, Gemini AI

Enhanced Differential Evolution and Particle Swarm Optimization Algorithms for IoT Applications

- Developed a hybrid **Differential Evolution and Particle Swarm Optimization (DE-PSO)** algorithm for complex IoT optimization problems.
- Improved convergence speed and solution accuracy compared to classical DE and PSO algorithms.
- Applied the algorithm to IoT-based multi-dimensional and non-linear optimization scenarios.

Background Removal Application

[GitHub](#)

- **Developed a full-stack SaaS background removal application**, enabling users to upload and process images seamlessly.
- **Integrated ClipDrop API** for accurate and efficient AI-based background removal with high-quality outputs.
- **Implemented authentication and payment system**, enabling secure user access and credit-based monetization.
- **Tools Used:** React.js, Node.js, Express.js, MongoDB, Tailwind CSS, Clerk, ClipDrop API

ACHIEVEMENTS AND CERTIFICATIONS

- Winner of **MetaMorph 1.0 Hackathon** organized by GNIT Kolkata, competing among 300+ registered teams.
- Solved 200+ **Data Structures and Algorithms** problems on LeetCode and GeeksforGeeks.
- Served as the **Lead of the college dev-club REPL Devs** and **Programming Head of the coding club HACKAUT**.
- Served as a Microsoft Learn Student Ambassador, contributing to technical workshops and peer learning initiatives.